

THE EXTREMAL GRAPH FOR $k(3, 4) = r(I_3, L_4) = 21$ IS NOT UNIQUE: THIRTEEN RIGID WITNESSES, AND A FORCED PALEY TOURNAMENT

DANIEL KIRTCHAKOV

ABSTRACT. The companion note [5] determines the Erdős–Rado oriented Ramsey number $k(3, 4) = r(I_3, L_4) = 21$ by an explicit $\{I_3, \text{TT}_4\}$ -free oriented graph W on 20 vertices together with a certified exhaustion at 21 vertices, and asks (Question 8.1) whether W is the unique such graph. We answer in the negative. There are at least thirteen pairwise non-isomorphic $\{I_3, \text{TT}_4\}$ -free oriented graphs on 20 vertices, and every one of the thirteen is *rigid* (trivial automorphism group), so the extremal configuration is far from a single symmetric object. We record the local structure they share as a short forcing lemma: in any $\{I_3, \text{TT}_4\}$ -free oriented graph a vertex has at most 7 non-neighbours, and a vertex with exactly 7 carries a copy of the Paley tournament $\text{QR}_7 = \text{Cay}(\mathbb{Z}_7, \{1, 2, 4\})$ as its non-neighbourhood. Finally the algebraic route falls short: the largest $\{I_3, \text{TT}_4\}$ -free tournament blow-up has only 14 vertices, six short of the extremal order, and the thirteen exhibited witnesses are rigid rather than Cayley. We close with the sharpened question of whether a QR_7 block is unavoidable in every witness.

1. THE QUESTION

An *oriented graph* is a loopless digraph with at most one arc between any two vertices. Write I_n for the independent set on n vertices and TT_m (equivalently L_m) for the transitive tournament on m vertices; call an oriented graph *free* if it contains neither I_3 nor TT_4 . For a vertex w let $N^+(w)$, $N^-(w)$, $I(w)$ be its out-neighbours, in-neighbours, and non-neighbours, and call $(p, q, s) = (|N^+(w)|, |N^-(w)|, |I(w)|)$ the *type* of w , so $p + q + s = |V| - 1$. The companion note [5] proves $k(3, 4) = r(I_3, L_4) = 21$: every oriented graph on 21 vertices contains an I_3 or a TT_4 , and the explicit 20-vertex graph W of its Table 1 is free. Its Question 8.1 asks:

Is W the unique free oriented graph on 20 vertices up to isomorphism? If not, what is the number of extremal graphs, and does any admit an algebraic description?

The first part has a clean negative answer.

Theorem 1.1. *There are at least thirteen pairwise non-isomorphic free oriented graphs on 20 vertices. Every one of the thirteen is rigid: its automorphism group is trivial. In particular W is not the unique 20-vertex extremal graph.*

The thirteen graphs are W itself together with twelve further free graphs; all thirteen are printed as arc lists with the artifacts. Each of the thirteen is in particular not vertex-transitive, since a

Date: August 11, 2026.

2020 Mathematics Subject Classification. Primary 05C55; Secondary 05C20, 05D10, 68V15.

Key words and phrases. Oriented graph, Ramsey number, transitive tournament, independent set, extremal graph, Paley tournament, rigidity, certified computation.

Computation and authorship. All witnesses, checkers, and enumerations in this note were produced by **Claude** (Anthropic), directed by the author, on a single Apple M4 laptop. The three shipped checkers import only the Python standard library and share no code with the search that produced the witnesses; the tournament and rigidity facts were re-established here by code written from scratch. This addendum uses no propositional solver: every claim is a finite check a reader can rerun.

Prior-art record. This note answers Question 8.1 of the companion determination [5] of $k(3, 4) = 21$. The value was absent from the literature before [5]: the Erdős problems entry [2] (#112) records no exact $k(n, m)$, and [4] remains the only paper on $r(I_m, L_n)$. A novelty sweep on the drafting date, August 11, 2026, found nothing on the number of extremal $\{I_3, \text{TT}_4\}$ -free graphs on 20 vertices, which is a fortiori new. *Draft of August 11, 2026.*

vertex-transitive graph on $n \geq 2$ vertices has $|\text{Aut}| \geq n$. We prove Theorem 1.1 by direct verification (§2); the local structure that the thirteen share is explained by the forcing lemma of §3; and the sense in which the algebraic construction falls short is made precise in §4.

2. THE WITNESSES

Table 1 lists the thirteen graphs by three isomorphism invariants: the number of arcs, the degree sequence of the non-adjacency graph $\overline{N}$ (whose edges are the non-adjacent pairs, so the $\overline{N}$ -degree of w is $s = |I(w)|$), and the number of vertices of type $s = 7$. By Lemma 3.1 below the last column equals the number of induced QR_7 blocks.

witness	arcs	$\overline{N}$ -degree sequence	$\#\{s=7\} = \#\text{QR}_7$ blocks
$w_1 = W$	126	$7^{11} 6^6 5^3$	11
w_2	125	$7^{12} 6^6 5^2$	12
w_3	128	$7^6 6^{12} 5^2$	6
w_4	125	$7^{12} 6^6 5^2$	12
w_5	126	$7^{11} 6^6 5^3$	11
w_6	123	$7^{14} 6^6$	14
w_7	126	$7^9 6^{10} 5^1$	9
w_8	126	$7^{12} 6^4 5^4$	12
w_9	124	$7^{14} 6^4 5^2$	14
w_{10}	127	$7^8 6^{10} 5^2$	8
w_{11}	124	$7^{13} 6^6 5^1$	13
w_{12}	125	$7^{13} 6^4 5^3$	13
w_{13}	125	$7^{11} 6^8 5^1$	11

TABLE 1. The thirteen free graphs on 20 vertices. Even this coarse fingerprint separates all thirteen: no two rows agree in all three columns (e.g. w_1 and w_5 share the first and second, but their full relation matrices differ; see the certification). Arc counts range over 123–128 and the number of QR_7 blocks over 6–14.

Each w_i was verified free by an exhaustive test over all $\binom{20}{3} = 1140$ triples (none independent) and all $\binom{20}{4} = 4845$ quadruples (none a TT_4 : a quadruple is a TT_4 exactly when its six pairs are all adjacent and its four in-set out-degrees are 0, 1, 2, 3). Rigidity, $|\text{Aut}(w_i)| = 1$, was certified two independent ways: Weisfeiler–Leman colour refinement reaches twenty distinct colours, so any automorphism fixes every vertex; and an explicit automorphism backtracker returns count 1. Pairwise non-isomorphism was decided by a canonical form read off the discrete refinement, and cross-checked by the invariant fingerprint of Table 1, which already separates all thirteen; no two are isomorphic.

Remark 2.1. The thirteen are only a lower bound on the extremal population. They were obtained by completing random partial orientations to free graphs and discarding isomorphic duplicates; in that sampling every completion that survived deduplication was a new isomorphism class, which suggests — but does not prove — that the family is large. Theorem 1.1 claims only what the thirteen exhibited certificates establish.

3. A FORCED PALEY BLOCK

The thirteen witnesses have no global symmetry, yet they share a rigid local feature, visible in the last column of Table 1: each carries several induced copies of the Paley tournament. This is forced by freeness alone.

Lemma 3.1 (QR_7 forcing). *Let D be a free oriented graph and w a vertex. Then $I(w)$ induces a tournament with no transitive subtournament on 4 vertices; consequently $|I(w)| \leq 7$, and if $|I(w)| = 7$ then $I(w)$ induces a tournament isomorphic to the Paley tournament $\text{QR}_7 = \text{Cay}(\mathbb{Z}_7, \{1, 2, 4\})$.*

Proof. If two vertices $x, y \in I(w)$ were non-adjacent, then $\{w, x, y\}$ would be an I_3 ; so every pair inside $I(w)$ is adjacent and $I(w)$ induces a tournament. A transitive subtournament on 4 vertices inside $I(w)$ is a TT_4 of D (for a tournament target, induced and subdigraph containment coincide), so $I(w)$ is TT_4 -free.

It remains to bound and identify a TT_4 -free tournament T . Every tournament on 8 vertices contains a transitive subtournament on 4 vertices — the tournament Ramsey value is $v(4) = 8$ — so $|V(T)| \leq 7$; and the unique tournament on 7 vertices with no transitive subtournament on 4 vertices is the quadratic-residue tournament QR_7 [6, 3]. Applying this to $T = I(w)$ gives $|I(w)| \leq 7$, with equality forcing $I(w) \cong \text{QR}_7$. $\square$

Both tournament inputs were re-established here from scratch, by exhaustive enumeration: there are exactly 240 labelled TT_4 -free tournaments on 7 vertices, and every one is isomorphic to QR_7 (so the class is unique; and $240 = 7!/|\text{Aut}(\text{QR}_7)|$ with $|\text{Aut}(\text{QR}_7)| = 21$), while there are none on 8 vertices. This matches the certified block inventory of [5, Prop. 3.3] and the classical characterisation [6].

Corollary 3.2. *In any free oriented graph every vertex has at most 7 non-neighbours; and a vertex has exactly 7 non-neighbours if and only if its non-neighbourhood is a copy of QR_7 .*

Corollary 3.2 is the endpoint $|I(w)| \leq 7$ that drives the counting bound of [5, §3]. It also explains the last column of Table 1: in each witness the number of induced QR_7 blocks equals the number of type- $s=7$ vertices, which was confirmed independently. Every one of the thirteen has between 6 and 14 such vertices, so QR_7 appears in each; whether it must appear in *every* free graph on 20 vertices is left open in §6.

4. THE ALGEBRAIC ROUTE FALLS SHORT

The natural algebraic construction from QR_7 is a blow-up, and it falls well short of the extremal order.

Proposition 4.1. *Let T be a tournament and $m \geq 1$, and let $T[I_m]$ be the lexicographic blow-up in which each vertex of T is replaced by an independent m -set. Then $T[I_m]$ is free if and only if $m \leq 2$ and T is TT_4 -free. Consequently the largest free graph of the form $T[I_m]$ is $\text{QR}_7[I_2]$ on 14 vertices, which gives only $k(3, 4) \geq 15$.*

Proof. In $T[I_m]$ two vertices are adjacent exactly when they lie in different groups (within a group there are no arcs; between groups the arc follows T , which is a tournament). Hence the underlying graph is complete multipartite with parts of size m , its independence number is m , and $T[I_m]$ is I_3 -free iff $m \leq 2$. A set of pairwise adjacent vertices meets each group at most once, so it projects injectively onto a subtournament of T and the projection is an isomorphism of induced sub-oriented-graphs; thus $T[I_m]$ contains a TT_4 iff T does. Combining the two, $T[I_m]$ is free iff $m \leq 2$ and T is TT_4 -free. The largest TT_4 -free tournament is QR_7 on 7 vertices (Lemma 3.1), so the largest such blow-up is $\text{QR}_7[I_2]$ on $2 \cdot 7 = 14$ vertices. $\square$

The bound $k(3, 4) \geq 15$ of Proposition 4.1 is six short of the truth 21: the extremal order is 20, but the best tournament blow-up reaches 14. The thirteen witnesses that reach the extremal order are rigid (Theorem 1.1), hence not vertex-transitive, hence not Cayley digraphs; and they are not blow-ups, since a proper blow-up has two vertices with identical neighbourhoods and every w_i has twenty distinct colour classes under refinement. In the language of [5, §8.1], where the extremal graph for $k(3, 3) = 9$ is a single circulant, the picture at $(3, 4)$ is the opposite: the extremal graph

is not unique, and the thirteen exhibited here are aperiodic and non-algebraic, with QR_7 surviving only as a forced local block.

5. CERTIFICATION

The thirteen witnesses (as arc lists) and three standard-library checkers are deposited with this note in the program’s public certificate repository *Certify* (github.com/05oz/certify; concept DOI [10.5281/zenodo.21799111](https://doi.org/10.5281/zenodo.21799111)), as Part J. From each witness’s arc list alone, `verify_witnesses.py` re-derives validity as an oriented graph, freeness over all $\binom{20}{3}$ triples and $\binom{20}{4}$ quadruples, rigidity (Weisfeiler–Leman refinement to twenty singleton colours, cross-checked by an explicit automorphism count), and pairwise non-isomorphism (a canonical form from the discrete refinement, cross-checked by the invariant fingerprint of Table 1); `verify_qr7_lemma.py` enumerates the labelled TT_4 -free tournaments on 7 and 8 vertices (240, all isomorphic to QR_7 ; and 0) and reports $|\text{Aut}(QR_7)| = 21$; and `blowup_bound.py` checks $QR_7[I_2]$ and $QR_7[I_3]$. All three checkers import only the Python standard library, share no code with the search that produced the witnesses, and return PASS.

6. THE SHARPENED QUESTION

Question 8.1 of [5] is answered: W is not unique, and the extremal count is at least 13. The thirteen exhibited here are rigid, hence none is vertex-transitive or Cayley; whether *some other* extremal graph might still be algebraic is not decided here. Three questions remain.

Question 6.1. Is a QR_7 block unavoidable? Equivalently, does every free oriented graph on 20 vertices have a vertex of type $s = 7$?

By Corollary 3.2 a vertex of type $s = 7$ is exactly a vertex whose non-neighbourhood is a QR_7 ; so Question 6.1 asks whether the propositional instance “free on 20 vertices with every vertex having at most 6 non-neighbours” is unsatisfiable. All thirteen exhibited witnesses carry between 6 and 14 such vertices, so QR_7 is present in each, but this is evidence, not a proof; the all-vertices- $s \leq 6$ instance was not decided in a short trial and a certified answer belongs to a dedicated computation.

Question 6.2. What is the exact number of free graphs on 20 vertices? The thirteen here are a lower bound; the sampling behaviour suggests a large family, but no count is certified.

Question 6.3. Can the counting slack of [5, §8.1] — the cap of 24 vertices against the true maximum of 20 — be explained structurally, for instance through the forced QR_7 and Bermond blocks? Such an explanation would bear on both the exact count and a human-readable upper bound.

REFERENCES

- [1] J.-C. Bermond, *Some Ramsey numbers for directed graphs*, Discrete Math. **9** (1974), 313–321.
- [2] T. F. Bloom, *Erdős Problem #112*, <https://www.erdosproblems.com/112>, accessed 2026-08-11.
- [3] P. Erdős and L. Moser, *On the representation of directed graphs as unions of orderings*, Magyar Tud. Akad. Mat. Kutató Int. Közl. **9** (1964), 125–132.
- [4] F. Ihringer, D. Rajendraprasad and T. Weinert, *New bounds on the Ramsey number $r(I_m, L_n)$* , Discrete Math. **344** (2021), no. 3, 112268. Also arXiv:1707.09556.
- [5] D. Kirtchakov, *An Erdős–Rado oriented Ramsey number determined: $k(3, 4) = r(I_3, L_4) = 21$, by explicit witness and certified exhaustion*, Certify (2026), version DOI [10.5281/zenodo.21890619](https://doi.org/10.5281/zenodo.21890619).
- [6] A. Sánchez-Flores, *On tournaments free of large transitive subtournaments*, Graphs Combin. **14** (1998), 181–200.

INDEPENDENT RESEARCHER (05oz); NO INSTITUTIONAL AFFILIATION. ORCID: [0009-0009-5213-4098](https://orcid.org/0009-0009-5213-4098).

Email address: daniel@halfounce.io

URL: <https://halfounce.io>